
APPENDIX B

LENNARD-JONES POTENTIALS AS DETERMINED FROM VISCOSITY DATA[†]

	Substance	$b_{0,\ddagger}$ cm ³ /g-mol	σ , Å	ϵ/k , K
Ar	Argon	56.08	3.542	93.3
He	Helium	20.95	2.551§	10.22
Kr	Krypton	61.62	3.655	178.9
Ne	Neon	28.30	2.820	32.8
Xe	Xenon	83.66	4.047	231.0
Air	Air	64.50	3.711	78.6
AsH ₃	Arsine	89.88	4.145	259.8
BCl ₃	Boron chloride	170.1	5.127	337.7
BF ₃	Boron fluoride	93.35	4.198	186.3
B(OCH ₃) ₃	Methyl borate	210.3	5.503	396.7
Br ₂	Bromine	100.1	4.296	507.9
CCl ₄	Carbon tetrachloride	265.5	5.947	322.7
CF ₄	Carbon tetrafluoride	127.9	4.662	134.0
CHCl ₃	Chloroform	197.5	5.389	340.2
CH ₂ Cl ₂	Methylene chloride	148.3	4.898	356.3
CH ₃ Br	Methyl bromide	88.14	4.118	449.2
CH ₃ Cl	Methyl chloride	92.31	4.182	350
CH ₃ OH	Methanol	60.17	3.626	481.8
CH ₄	Methane	66.98	3.758	148.6
CO	Carbon monoxide	63.41	3.690	91.7
COS	Carbonyl sulfide	88.91	4.130	336.0
CO ₂	Carbon dioxide	77.25	3.941	195.2
CS ₂	Carbon disulfide	113.7	4.483	467
C ₂ H ₂	Acetylene	82.79	4.033	231.8
C ₂ H ₄	Ethylene	91.06	4.163	224.7
C ₂ H ₆	Ethane	110.7	4.443	215.7
C ₂ H ₅ Cl	Ethyl chloride	148.3	4.898	300
C ₂ H ₅ OH	Ethanol	117.3	4.530	362.6
C ₂ N ₂	Cyanogen	104.7	4.361	348.6
CH ₃ OCH ₃	Methyl ether	100.9	4.307	395.0
CH ₂ CHCH ₃	Propylene	129.2	4.678	298.9

B.1

Substance		$b_0^{\ddagger}$ cm ³ /g-mol	σ , Å	ϵ/k , K
CH ₃ CCH	Methylacetylene	136.2	4.761	251.8
C ₃ H ₆	Cyclopropane	140.2	4.807	248.9
C ₃ H ₈	Propane	169.2	5.118	237.1
<i>n</i> -C ₃ H ₇ OH	<i>n</i> -Propyl alcohol	118.8	4.549	576.7
CH ₃ COCH ₃	Acetone	122.8	4.600	560.2
CH ₃ COOCH ₃	Methyl acetate	151.8	4.936	469.8
<i>n</i> -C ₄ H ₁₀	<i>n</i> -Butane	130.0	4.687	531.4
<i>iso</i> -C ₄ H ₁₀	Isobutane	185.6	5.278	330.1
C ₂ H ₅ OC ₂ H ₅	Ethyl ether	231.0	5.678	313.8
CH ₃ COOC ₂ H ₅	Ethyl acetate	178.0	5.205	521.3
<i>n</i> -C ₅ H ₁₂	<i>n</i> -Pentane	244.2	5.784	341.1
C(CH ₃) ₄	2,2-Dimethylpropane	340.9	6.464	193.4
C ₆ H ₆	Benzene	193.2	5.349	412.3
C ₆ H ₁₂	Cyclohexane	298.2	6.182	297.1
<i>n</i> -C ₆ H ₁₄	<i>n</i> -Hexane	265.7	5.949	399.3
Cl ₂	Chlorine	94.65	4.217	316.0
F ₂	Fluorine	47.75	3.357	112.6
HBr	Hydrogen bromide	47.58	3.353	449
HCN	Hydrogen cyanide	60.37	3.630	569.1
HCl	Hydrogen chloride	46.98	3.339	344.7
HF	Hydrogen fluoride	39.37	3.148	330
HI	Hydrogen iodide	94.24	4.211	288.7
H ₂	Hydrogen	28.51	2.827	59.7
H ₂ O	Water	23.25	2.641	809.1
H ₂ O ₂	Hydrogen peroxide	93.24	4.196	289.3
H ₂ S	Hydrogen sulfide	60.02	3.623	301.1
Hg	Mercury	33.03	2.969	750
HgBr ₂	Mercuric bromide	165.5	5.080	686.2
HgCl ₂	Mercuric chloride	118.9	4.550	750
HgI ₂	Mercuric iodide	224.6	5.625	695.6
I ₂	Iodine	173.4	5.160	474.2
NH ₃	Ammonia	30.78	2.900	558.3
NO	Nitric oxide	53.74	3.492	116.7
NOCl	Nitrosyl chloride	87.75	4.112	395.3
N ₂	Nitrogen	69.14	3.798	71.4
N ₂ O	Nitrous oxide	70.80	3.828	232.4
O ₂	Oxygen	52.60	3.467	106.7
PH ₃	Phosphine	79.63	3.981	251.5
SF ₆	Sulfur hexafluoride	170.2	5.128	222.1
SO ₂	Sulfur dioxide	87.75	4.112	335.4
SiF ₄	Silicon tetrafluoride	146.7	4.880	171.9
SiH ₄	Silicon hydride	85.97	4.084	207.6
SnBr ₄	Stannic bromide	329.0	6.388	563.7
UF ₆	Uranium hexafluoride	268.1	5.967	236.8

[†]R. A. Svehla, *NASA Tech. Rep.* R-132, Lewis Research Center, Cleveland, Ohio, 1962.

[‡] $b_0 = \frac{2}{3}\pi N_0 \sigma^3$, where N_0 is Avogadro's number.

[§]The parameter σ was determined by quantum-mechanical formulas.